GRÖBNER BASES AND CONVEX POLYTOPES

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Contents

1. Lesson 1: Gröbner bases

1.1. **Introduction.** We follow

- 'Gröbner bases and convex polytopes' by Sturmfels.
- 'Ideals, Varieties, and Algorithms' by Cox, Little, and O'Shea.

1.2. **Definitions and notation.** Let k be a field (I think most of this work in general, but let's assume algebraically closed just in case). We write the polynomial ring over k in n variables as

$$k[x_1, \dots, x_n]$$

Definition 1.1. The ideal I generated by $f_1, \dots, f_m \in k[x_1, \dots, x_n]$ is defined as

$$I = (f_1, \dots, f_m) = \{h_1 f_1 + \cdots h_m f_m \mid h_i \in k[x_1, \dots, x_n]\}$$

Exercise 1.1. Show that $I = (f_1, \dots, f_m)$ really is an ideal.

Note that $I = (f_1, \dots, f_m)$ is a *finitely generated* ideal. We call $\{f_1, \dots, f_m\}$ a basis of I. If $I = (f)$ then we call it *principal.*

In general ideals (when I am thinking about any rings, not just polynomial rings), do not have to be finitely generated. It turns out that in the polynomial ring case, every ideal is finitely generated, which is very nice. This is Hilbert's Basis Theorem, which we will state and prove later

Remark 1.1. Why do people case so much about ideals in polynomial rings? This is because, if you are given and ideal $I = (f_1, \dots, f_m) \subset k[x_1, \dots, x_n]$, then we can consider a geometry object (a *variety*) defined as

$$V(I) = \{(a_1, \dots, a_n) \in \mathbb{R}^n \mid f_1(a_1, \dots, a_n) = \cdots = f_m(a_1, \dots, a_n) = 0\}$$

i.e. the zero set of the polynomials. So, we can think that if we want to study geometric properties of $V(I)$ we want to write I in a nice way, hence we need to study how to write a nice generating set of the ideal.

Note that you can go back as well

$$I(V) = \{f \in k[x_1, \dots, x_n] \mid f(a_1, \dots, a_n) = 0 \text{ for all } (a_1, \dots, a_n) \in V\}$$

Unfortunately, $I(V(I)) \neq I$, but at least we have $I \subset I(V(I))$. To see this look at $I = (x^2) \subset k[x]$. We have

$$I(V(I)) = I(\{x = 0\}) = (x)$$

A reasonable question is: since every ideal in a polynomial ring is finitely generated, can we build a basis? This is the question that **Gröbner bases** (a fundamental building block in computational algebraic geometry) answers.

1.3. **Polynomials in $k[x]$.** Let's think about polynomials in one variable for a second. Life is easy.

Definition 1.2. Let $f \in k[x]$ be a non-zero polynomial. Then we can write

$$f = a_0 + a_1 x + \cdots + a_m x^m$$

We say $m = \deg(f)$ (the degree of the polynomial) and $a_m = \mathrm{LT}(f)$ (the leading term of the polynomial).

Proposition 1.1. *Let $g \in k[x]$ be a non-zero polynomial. Then every $f \in k[x]$ can be written uniquely in the form*

$$f = gq + r$$

where $q, r \in k[x]$ with $\deg(r) < \deg(g)$ or $r = 0$.

Proof. Let $q = 0$, set $r = f$. If $\deg(f) < \deg(g)$ or $r = 0$ we are done. Otherwise set

$$q' = q + \frac{\mathrm{LT}(r)}{\mathrm{LT}(g)} \quad r' = r - \frac{\mathrm{LT}(r)}{\mathrm{LT}(g)} g$$

Then clearly $f = q'g + r'$. This process (inductively) reduces the degree of r, so we have shown existence.

For the uniqueness, suppose

$$f = qg + r = q'g + r'$$

with $\deg(r) < \deg(g)$ or $r = 0$ and $\deg(r') < \deg(g)$ or $r' = 0$. If $r \neq r'$, then $\deg(r - r') < \deg(g)$. This also implies $q \neq q'$, leading to a contradiction

$$\deg(r - r') = \deg((g - g')q) = \deg(q - q') + \deg(g) \geq \deg(g)$$

□

Corollary 1.1. *Every ideal I in $k[x]$ is principal. Furthermore, if $I = (f)$ for some $f \in k[x]$ then f is unique up to multiplication by a non-zero scalar in k.*

Proof. If $I = (0)$ we are done.

Otherwise, let $f \in I$ be a non-zero polynomial of minumum degree. We claim $(f) = I$. Let $g \in I$. Then $g = qf + r$ for $r = 0$ or $\deg(r) < \deg(f)$. However, $r \in I$, so by minimality of degree of f we must have $r = 0$. So $I = (f)$.

Assume $(f) = (g)$, then $f = hg$ and $g = kf$ for some $h, k \in k[x]$. Therefore $\deg(g) = \deg(f)$ so they differ by multiplication by a non-zero constant. □

1.4. **Polynomials in $k[x_1, \ldots, x_n]$.** The aim of this section is to be able to generalise the results from the previous section to an arbitrary polynomial ring $k[x_1, \ldots, x_n]$. We will see that the nice picture from $k[x]$ is not fully recovered.

First we need a way to order monomials in multiple variables. Recall that monomials are things like $x_1^{\alpha_1} \cdots x_n^{\alpha_n}$ for $(\alpha_1, \ldots, \alpha_n) \in \mathbb{Z}^n_{\geq 0}$. We have a 1-1 correspondence between monomials in $k[x_1, \ldots, x_n]$ and $\mathbb{Z}^n_{\geq 0}$.

Definition 1.3. A **total order** on $\mathbb{Z}^n_{\geq 0}$ is defined as a binary relation $>$ such that

- For all $\alpha, \beta, \gamma \in \mathbb{Z}^n_{\geq 0}$, $\alpha > \beta$ and $\beta > \gamma$ implies $\alpha > \gamma$;
- For all $\alpha, \beta \in \mathbb{Z}^n_{\geq 0}$ we have on of the following: $\alpha > \beta$ or $\alpha = \beta$ or $\beta > \alpha$.

We can use a total order on $\mathbb{Z}^n_{\geq 0}$ to define a *monomial order* on monomials in $k[x_1, \ldots, x_n]$.

Definition 1.4. Let $>$ be a total order on $\mathbb{Z}^n_{\geq 0}$ such that

- If $\alpha > \beta$ then $\alpha + \gamma > \beta + \gamma$ for all $\gamma \in \mathbb{Z}^n_{\geq 0}$,
- For any non-empty subset of $\mathbb{Z}^n_{\geq}$ there exists a smallest element with respect to $>$

Then a **monomial order** is a binary relation $\succ$ on monomials in $k[x_1, \ldots, x_n]$ such that $\mathbf{x}^\alpha \succ \mathbf{x}^\beta$ if and only if $\alpha > \beta$.

Example 1.1.

- (Lexicographic order) $\alpha >_{\text{lex}} \beta$ if the left-most non-zero entry of $\alpha - \beta$ is positive. For example if $x_1 > x_2 > x_3$, then $x_1^2 x_2^3 \succ_{\text{lex}} x_1^2 x_2^2 x_3^5$ because the left-most non-zero entry of $(2, 3, 0) - (2, 2, 5)$ is positive.
- (Graded lexicographic order) $\alpha >_{\text{grlex}} \beta$ if $\sum_i \alpha_i > \sum_i \beta_i$ or $\sum_i \alpha_i = \sum_i \beta_i$ and $\alpha >_{\text{lex}} \beta$. Now $x_1^2 x_2^2 x_3^5 \succ_{\text{grlex}} x_1^2 x_2^3$ because $\sum_i \alpha_i = 5 < 9 = \sum_i \beta_i$.

Definition 1.5. Let $f \in k[x_1, \ldots, x_n]$ and $\succ$ be a monomial order. We can write $LT(f)$ as the leading term of f with respect to $\succ$, i.e. the term of f with the largest monomial with respect to $\succ$.

Theorem 1.1. *Let $\succ$ be a monomial order on $k[x_1, \ldots, x_n]$ and $f_1, \ldots, f_s \in k[x_1, \ldots, x_n]$. Then every $f \in k[x_1, \ldots, x_n]$ can be written in the form*

$$f = h_1 f_1 + \cdots + h_s f_s + r$$

where $h_i, r \in k[x_1, \ldots, x_n]$ and no term of r is divisible by any of the leading terms $LT(f_i)$.

This is almost like the division algorithm in $k[x]$, but we have a problem: the representation of f is not unique. For example, let $f_1 = x_1$ and $f_2 = x_1 + x_2$. Then we can write x_1 as

$$x_1 = 1 \cdot f_1 + 0 \cdot f_2 + 0$$

or as

$$x_1 = 2 \cdot f_1 - 1 \cdot f_2 + x_2$$

We won't be able to recover from the non-uniqueness of the h_i's, but we are able to do something about the non-uniqueness of r. This is what Gröbner bases are for.

1.5. **Gröbner bases.**

Definition 1.6. Let I be an ideal in $k[x_1, \dots, x_n]$. If

$$I = (x_\alpha \mid \alpha \in A)$$

for some $A \subset \mathbb{Z}^n_{\geq 0}$, then we call I a **monomial ideal**.

Theorem 1.2 (Dickson's Lemma). *Every monomial ideal in $k[x_1, \dots, x_n]$ is finitely generated.*

Definition 1.7. Let I be an ideal in $k[x_1, \dots, x_n]$ and $\succ$ a monomial order. The **initial ideal** of I with respect to $\succ$ is the monomial ideal generated by the leading terms of the elements of I, i.e.

$$\text{in}_\succ(I) = (LT(f) \mid f \in I)$$

Example 1.2. Unfortunately, the initial ideal of an ideal I is not necessarily finitely generated by the leading terms of a finite generating set of I. For example, let $I = (x_1, x_1 + x_2)$ with $x_1 \succ x_2$. Then $(\text{LT}(x_1), \text{LT}(x_1 + x_2)) = (x_1)$, which is strictly contained in $\text{in}_\succ(I)$ since it does not contain x_2.

Theorem 1.3 (Hilbert's Basis Theorem). *Every ideal in $k[x_1, \dots, x_n]$ is finitely generated.*

Proof. Let I be an ideal in $k[x_1, \dots, x_n]$ and $\succ$ a monomial order. Then $\text{in}_\succ(I)$ is a monomial ideal, so by Dickson's Lemma it is finitely generated, so there exist $f_1, \dots, f_m \in I$ such that $\text{in}_\succ(I) = (LT(f_1), \dots, LT(f_m))$. We claim that $I = (f_1, \dots, f_m)$.

Let $f \in I$. We can write

$$f = h_1 f_1 + \cdots + h_m f_m + r$$

where no term of r is divisible by any of the leading terms $LT(f_i)$. If $r = 0$ we are done. Otherwise, since $f, h_i, f_i \in I$, we have that $r = f - (h_1 f_1 + \cdots + h_m f_m) \in I$. This implies that $LT(r) \in \text{in}_\succ(I)$, so there exists some i such that $LT(f_i)$ divides $LT(r)$, which is a contradiction. □

So, not only we have shown that every ideal in $k[x_1, \dots, x_n]$ is finitely generated, but we have also shown that we can find a finite generating set of I such that the leading terms of the generators generate the initial ideal of I. This motivates the following definition.

Definition 1.8. Let I be an ideal in $k[x_1, \dots, x_n]$ and $\succ$ a monomial order. A finite subset $G = \{g_1, \dots, g_t\}$ of I is called a **Gröbner basis** of I with respect to $\succ$ if the ideal generated by the leading terms of the elements of G is equal to the ideal generated by the leading terms of the elements of I, i.e.

$$(LT(g_1), \dots, LT(g_t)) = (LT(f) \mid f \in I)$$

So, in the theorem above we have shown that every ideal in $k[x_1, \dots, x_n]$ has a Gröbner basis and that a Gröbner basis is a generating set of the ideal. However, a Gröbner basis does not have to be unique, and it might have many redundancies.

Gröbner bases are very useful because they allow us to recover the uniqueness of r that we had lost in our new division algorithm.

Proposition 1.2. *Let $f \in k[x_1, \dots, x_n]$. Let $I = (g_1, \dots, g_t)$ be a Gröbner basis of I with respect to $\succ$. Then, $f \in I$ if and only if the remainder on division of f by $g_1, \dots, g_t$ is zero.*

The h_i's are still not unique, but this recovers a lot of the unhappiness we were feeling before in our division algorithm.

1.6. **Computing Gröbner bases.** Computing Gröbner bases is a very important problem in computational algebraic geometry. The most famous algorithm to compute Gröbner bases is Buchberger's algorithm, which underpins a lot of different computer algebra computations. We won't go into the details of the algorithm, but we will just state it and give some intuition on how it works.

Definition 1.9. Let $f, g \in k[x_1, \ldots, x_n]$ and $\succ$ a monomial order. The **S-polynomial** of f and g is defined as

$$S(f,g) = \frac{\operatorname{lcm}(LT(f), LT(g))}{LT(f)} f - \frac{\operatorname{lcm}(LT(f), LT(g))}{LT(g)} g$$

The S-polynomial is a way to measure how much f and g fail to be a Gröbner basis.

Theorem 1.4 (Buchberger's Criterion)**.** *Let I be an ideal in $k[x_1, \ldots, x_n]$ and $\succ$ a monomial order. A finite subset $G = \{g_1, \ldots, g_t\}$ of I is a Gröbner basis of I with respect to $\succ$ if and only if the remainder on division of $S(g_i, g_j)$ by G is zero for all $i \neq j$.*

Buchberger's algorithm is an iterative algorithm that starts with a generating set of I and then computes the S-polynomials of the generators. If the remainder on division of any S-polynomial is not zero, then we add it to our generating set and repeat the process until we get a Gröbner basis. This process terminates since we create a strictly increasing sequence of monomial ideals, and there are no infinite strictly increasing sequences of monomial ideals (this is a consequence of Dickson's Lemma).

Definition 1.10. A Gröbner basis G of an ideal I is called **reduced** if for all $g \in G$ we have that the coefficient of $LT(g)$ is 1 and no term of g is divisible by any of the leading terms of the other elements of G.

Theorem 1.5. *Every ideal in $k[x_1, \ldots, x_n]$ has a unique reduced Gröbner basis with respect to a given monomial order.*

Note that we can also define a more universal object, called a *universal Gröbner basis*, which is a finite subset of I that is a Gröbner basis of I with respect to every monomial order. However, we won't go into the details of this object, but it is worth mentioning that it exists and it is unique (the fact that it exists is a consequence of the fact that there are only finitely many initial ideals of I).